

Comparison of Tabular Q-Learning and Deep Q-Networks (DQN) for Snake

Varun Sri Teja

July 9, 2025

1 Introduction

Reinforcement learning (RL) has been successfully applied to classic games such as Snake. In this week, we compare two RL approaches: Tabular Q-learning and Deep Q-Networks (DQN). We analyze their performance and learning characteristics on the Snake game environment.

2 Tabular Q-Learning

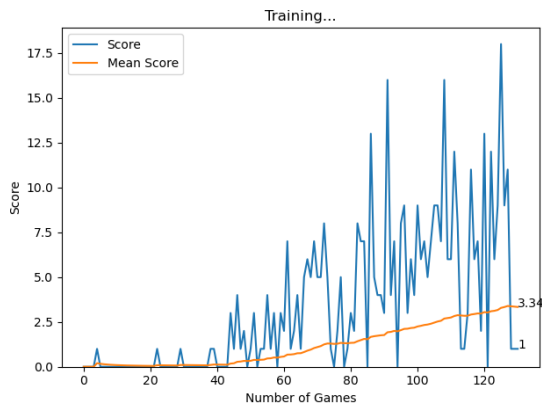
Tabular Q-learning is a model-free RL algorithm that maintains a table of Q-values for each state-action pair. It is suitable for environments with small, discrete state spaces. In our implementation, the state was discretized, and the agent learned optimal actions through repeated interactions.

3 Deep Q-Network (DQN)

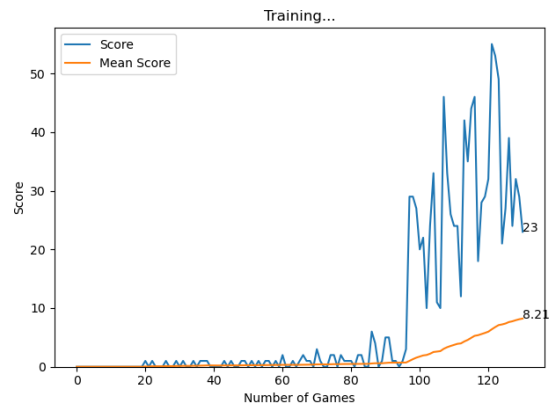
DQN extends Q-learning by using a neural network to approximate the Q-value function, enabling it to handle high-dimensional or continuous state spaces. In our project, DQN was trained using a feature-based state representation.

4 Comparison

DQN generally achieves higher scores and learns faster due to its ability to generalize from features, while Tabular Q-learning is limited by the discretization of the state space.



(a) Tabular Q-learning



(b) DQN